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RESEARCH-ARTICLE

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Attention Based Echo State Network: A Novel Approach for Fault Prognosis

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ABSTRACT
Recurrent neural networks (RNNs) are widely studied in recent years, since RNNs are capable of modeling the significant nonlinear dynamical systems. Echo state network (ESN) is a novel type of RNN with an interconnected reservoir to model temporal dynamics of complex sequential information. In this paper, a novel ESN structure is developed and employed to conduct fault prognosis. Fault prognosis is vital in predictive maintenance, which is a prevalent research area that mainly concentrates on predicting the remaining useful life of a machine and reducing the machine's downtime. Attention model is integrated to a typical ESN and thus different importance levels of different input elements can be adaptively treated. To further enhance the generalization of the prediction model, genetic algorithm is applied to adaptively optimize the parameters of the attention-based ESN. The proposed prognostic approach is verified on the NASA's turbofan benchmark dataset. Experimental results show that the attention-based ESN can not only achieve superior prediction accuracy but also obtain substantial improvement on stability.

CCS Concepts
• Computing methodologies →Machine learning →Supervised learning by regression• Computing methodologies →Neural network.

Keywords
Echo state network; attention mechanism; fault prognosis; genetic algorithm; remaining useful life.

1. INTRODUCTION
In recent years, fault prognosis, which is very cost effective and plays an important role in predictive maintenance, has attracted more and more attention in study area. Prognostics mainly focus on predicting the remaining useful life (RUL) of a degrading system. A reliable RUL prediction provides valuable information for condition based maintenance, thus reducing downtime of equipment and improving business efficiency, such as productivity and elimination of malfunctions [1]. Basically, prognostic approaches can be categorized as three classes, namely model based, data-driven, and hybrid [2]. Model-based approaches utilize an explicit mathematical model of the degradation process to predict the future evolution of the degradation state and RUL of the system [3]. Data-driven approaches are mostly based on statistical and machine learning techniques that contribute to learning trends from the monitoring data and discovering the system's behavior. Finally, hybrid approaches are the combination of model-based and data-driven approaches that benefit from leveraging the strength of the both categories [4].

Driven by the rapid development of Internet of Things and industrial big data, the data-driven approaches are drawing more attention nowadays. Based on massive monitoring data, data-driven methods dedicate to mining the degradation trends from the point of data mapping, in which way the physical model of system can be freed, especially when the system is too complex to modelling completely. Typically, the data-driven approaches include but not limit to artificial neural network [5], support vector regression (SVR) [6], recurrent neural networks (RNNs) [7, 8], and Bayesian models [9], et al. Specifically, RNN has the ability to model significant nonlinear dynamical systems, which is appropriate for handling the prediction problems related to time series. It is worth mentioning that some RNN based methods, such as long short-term memory (LSTM) and echo state network (ESN), have been successfully adopted to predict the RULs of different systems. In contrast to LSTM method, ESN holds a distinguishable simple learning procedure, which makes the RNN training computationally more efficient [10]. There are some variants of RNNs with slightly different designs such as ESN, which combats the vanishing gradients problem though regression-based training without using gradients. However, the ESNs are not designed to explore the potential different characteristics of the input elements.

Attention mechanism which selectively put emphasis on different parts of the input data has been demonstrated to be effective for many applications [11, 12]. In a recent work, an Element-wise-Attention Gate based RNN block is designed by adding attentiveness to the neurons in general RNN, the effectiveness of which is demonstrated on the action recognition tasks [13]. Inspired by these, in this paper an attention based ESN (Atten-ESN) is developed as a novel data-driven approach for RUL prediction. The proposed Atten-ESN is designed to modulate the input elements via adding attentiveness to a standard ESN, in which way the ability of modelling the sequential data could be enhanced. Furthermore, to improve the accuracy and

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generalization ability of the RUL prediction, genetic algorithm is adopted to optimize the structure parameters of the ESN. Experiment results on turbofan benchmark dataset shows the effectiveness of the proposed approach.

2. METHODOLOGY

2.1 Ridge Regression Based Echo State Network

An ESN mainly comprises two basic parts, one is a discrete-time, recurrent, and randomly connected dynamic hidden layer, called reservoir, and the other is a linear readout layer which maps the reservoir states to the predicted output, the structure of an ESN is shown in Fig.1. In this work, a supervised ESN training based on ridge regression is considered. The updating processes of reservoir states and network output are given as follows:

$$h(t+1) = (1-\lambda)\phi\big(W_{in}u(t+1) + Wh(t) + W_{back}\hat{y}(t) + \delta\big) + \lambda h(t) \quad (1)$$

$$y(t+1) = W_{out}\lfloor h(t+1); u(t+1)\rfloor \quad (2)$$

where δ is a small *i.i.d* noise term, $u(t)$ is the observed input fed to the reservoir at time t, $h(t)$ is reservoir states at time t, y and $\hat{y}$ are predicted readout and the target readout respectively. W denotes the reservoir weight matrix, W_{in} and W_{back} are weight matrices of the input and output feedback. All of the three weight matrices related to the reservoir are randomly generated from a specific probability distribution. W_{out} denotes the readout weights matrix which is identified by ridge regression in training phase. $\phi(\bullet)$ denotes the activation function of reservoir neurons, and $\lambda \in [0,1]$ is the leaky rate. After the training process, reservoir states are updated as

$$h(t+1) = (1-\lambda)\phi\big(W_{in}u(t+1) + Wh(t) + W_{back}y(t) + \delta\big) + \lambda h(t) \quad (3)$$

Ridge regression is adopted to mitigate the ill-condition problem via regularization strategy [14]. Therefore, the readout weights can be calculated as

$$W_{out} = \left(H^T H + \xi I\right)^{-1} H^T \hat{Y} \quad (4)$$

where $H = (H(1), \mathrm{H}(2), ..., \mathrm{H}(\mathrm{T}))^T$, and $H(\mathrm{t}) = [h(\mathrm{t}); \mathrm{u}(\mathrm{t})]$, $t = 1, 2, ..., T$. ξ is the regularization coefficient. $\hat{Y}$ is the target vector corresponding to input teacher-forced signals. In addition, to reduce the effect of random initialization of those aforementioned weights, the echo state property should be satisfied. For a detailed description of the ESN theory and application, the interested readers can refer to [10].

From the designs of ESN, we note that only the output weights in the network need to be trained. Typically, regression-based methods are applied for the training of ESN rather than gradient-based methods, which leads to a simpler learning procedure. However, the controlling of the sequence takes the input $u(\mathrm{t})$ as a whole without adaptively regarding different elements of the input. Thus the differences among different elements of the input could be overlooked. To address this problem, attention mechanism is introduced to the ESN, the details of which is presented in the next section.

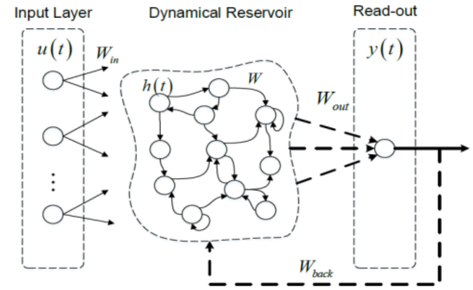

Figure 1. Structure of Echo State Network.

2.2 Attention Based Echo State Network

Regarding to the input layer of an ESN, we note that different elements of the input represent different features. In order to handle different features information of the input data, attention mechanism is added to the input layer and different elements of the input flow can be adaptively treated. The response of the attentiveness is a vector $a(\mathrm{t})$ with the same dimension as the input $u(\mathrm{t})$ of the ESN, which is updated as

$$\mathrm{a(t)} = \varphi(\mathrm{W}_{ua}\mathrm{u(t)} + \mathrm{W}_{ha}\mathrm{h(t-1)} + \mathrm{b}_a) \quad (5)$$

where $\varphi(\bullet)$ denotes the activation function and sigmoid is adopted in this work, i.e., $\varphi(\mathrm{x}) = 1/(1+\mathrm{e}^{-x})$. W_{ua} denotes the weights matrix related to the input and W_{ha} denotes the weight matrix related to the reservoir states. b_a denotes the bias vector. The input $u(\mathrm{t})$ at time t and the hidden state $\mathrm{h(t-1)}$ at time $t-1$ are taken to determine the levels of importance of each feature or element of the input $u(\mathrm{t})$.

Hence, the original input is modulated by the attention response and an updated $\hat{u}(\mathrm{t})$ is given as

$$\hat{u}(\mathrm{t}) = \mathrm{a(t)} \odot \mathrm{u(t)} \quad (6)$$

where $\odot$ denotes element-wise multiplication. The recursive computations of activations of the ESN block are then based on the updated input $\hat{u}(\mathrm{t})$, rather than the original input $u(\mathrm{t})$, as illustrated in Fig.2.

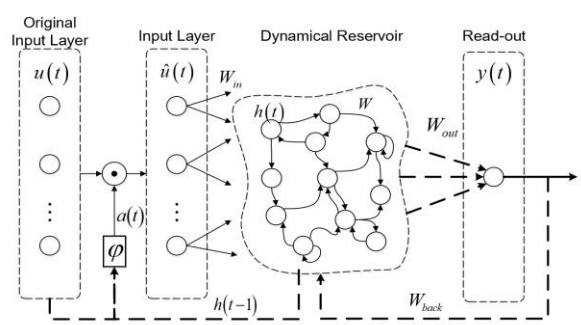

Figure 2. Structure of Attention Based Echo State Network.

For a standard ESN model with attention mechanism (denoted as Atten-ESN), the reservoir state $h(t+1)$ at time $t+1$ derived from (1) can be replaced by

$$h(t+1)=(1-\lambda)\phi\big(W_{in}\hat{u}(t+1)+Wh(t)+W_{back}\hat{y}(t)+\delta\big)+\lambda h(t) \quad (7)$$

Typically, most attention designs use Softmax as the activation function and thus the attention values can sum to one. However as explained in [13], once the sum-to-one constraint is held, the attention response values can be affected by the changes of other elements' response values, even when the importance levels of this element are the same over consecutive time series. In this work, the sum-to-one constraint is relaxed by using the sigmoid activation function, which limits the response values varying from 0 to 1.

2.3 Proposed RUL Prediction Approach

This paper proposes a novel data-driven approach for RUL prediction based on the developed Atten-ESN. To begin with, feature selection is performed to pick up some suitable features that benefit the prediction task. Then based on the features selected from the original data, the Atten-ESN model is utilized to conduct the RUL prediction. Moreover, aiming to improve the accuracy of the prediction model, an evolutionary algorithm called genetic algorithm (GA) is employed to optimize the hyper-parameters of the Atten-ESN model. The procedure of the optimization algorithm is depicted in Table 1.

Table 1. The parameters optimization algorithm

Algorithm: GA for parameters optimization of Atten-ESN

1. **Input:** Training data, crossover rate **Pc**, mutation rate **Pm**, maximum iteration times **T**.
2. **Initialization:** Initialize the reservoir size **S**, the spectral radius **R**, the input weight matrix W_{in}, the reservoir matrix W, the output weight matrix W_{out}, the feedback matrix W_{back}, the attention related matrix (W_{ua}, W_{ha}), and the population size.
3. **Optimization:** (iteration index **i** = 0)
4. **While i < T**
5. **Fitness evaluation:** cross validation is conducted and in each fold, the Atten-ESN is used to predict RUL, then the losses are computed and cumulated as the fitness value.
6. **Selection operator:** tournament strategy is adopted.
7. **Crossover:** the area of crossover is decided by two random numbers, then partial slices of two parents are exchanged.
8. **Mutation:** mutating towards the elite with the probability **Pm**.
9. **Population updating:** Preserve a certain number of superior chromosomes
10. **Judgement:** Decide whether to stop earlier or not, **i** ++.
11. **end While**
12. **Output:** The optimal parameters which including the reservoir size **S**, the spectral radius **R**, the ESN related weights and the attention related weights.

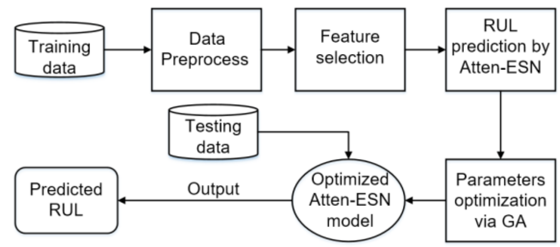

Figure 3. Overview of the proposed RUL prediction approach

The overview of the proposed RUL prediction approach based on the Atten-ESN is presented in Fig.3. As shown in the figure, the optimized Atten-ESN model is obtained after parameters optimization via GA, then the RUL prediction task of the test samples are performed with the optimized model. It is worth mentioning that the weight matrices related to attention in (5) are optimized via GA as well, while in most other attention designs the weight parameters are updated by gradient-based methods.

3. EXPERIMENTAL VALIDATIONS

To validate the efficiency of the developed prognostic approach, RUL prediction task is conducted on the turbofan engine degradation problem. The benchmarking turbofan dataset is available in NASA's data repository [15].

3.1 Dataset Description and Preprocessing

Turbofan engine degradation dataset comprises four sub-datasets and each sub-dataset includes one training set and one test set, both of which contain a certain number of engines. Degradation behaviors of each engine in training set are represented by given trajectories corresponding to multivariate temporal data points. In detail, each data point is a snapshot including not only 21 signals recording the component operation of the sensory measurement, but also operation settings, engine IDs and operational cycle. Table 2 shows some basic information of one sub-dataset, i.e., FD001, which is considered in this work. The data collected from various parts of the system record effects of degradations on sensor measurements such as pressure, temperature, speed, etc. For more details the readers can refer to [16].

Table 2. Basic information of C-MAPSS dataset FD001

Dataset	FD001
Fault mode	1
Operational condition	1
Engines in train set	100
Engines in test set	100
Minimum lifespan (cycles)	128
Maximum lifespan (cycles)	362

In addition, some preprocessing strategies have been adopted to efficiently handle the complexity of the dataset. Firstly, the sensors with small variance are removed and some redundant signals are further removed via correlation analysis, in which way some suitable features for the prediction task are attained. Then, the signal values from each sensor are normalized by min-max scaling and each sensor holds a value range between 0 and 1. Finally, time window is applied to compress the data, as such the mean value and the differential value per dimension are extracted from each window. In this case, the width of time window is identified by trial and error, which is set as 10 time cycles.

3.2 Evaluation Metrics

Two different evaluation metrics which reflect the quantities of interest in a practical application, are used to assess the performance of the proposed approach. The first is the mean square error (MSE), which is employed to measure the accuracy of RUL prediction and is defined as follow:

$$MSE = \frac{1}{N}\sum_{n=1}^{N}(RUL_i - RUL_{true})^2 \tag{8}$$

where N is the number of engines in the test set, RUL_i and RUL_{true} denote the predicted RUL and the ground truth RUL respectively. Additionally, the scoring metric [16] is adopted and the score S is defined as

$$S = \begin{cases} \sum_{i=1}^{N}\left(e^{-d_i/13} - 1\right), & \text{for } d_i < 0 \\ \sum_{i=1}^{N}\left(e^{d_i/10} - 1\right), & \text{for } d_i \geq 0 \end{cases} \tag{9}$$

where $d_i = RUL_i - RUL_{true}$ is the error term between predicted RUL and ground truth RUL. Note that the late predictions are more heavily penalized than the early predictions, and the prediction approaches with better performance will yield lower score values.

3.3 RUL Prediction with Atten-ESN

Since the preprocessed data are prepared, the developed Atten-ESN approach is ready for RUL prediction. In training procedure, the GA is adopted for parameters optimization, the included parameters such as reservoir size, spectral radius, input weight, output feedback scale and the attention related weights. In addition, a five-fold cross validation is applied and every cross validation runs two times to relieve the overfitting problem. In the GA-based optimization, the crossover rate is 0.8, the mutation rate is 0.5, and the population size is set as 600. Moreover, connectivity ratio of the reservoir units is set as 0.05 since sparse connectivity contributes to generating a rich variety of dynamics in reservoir. Ridge regression is conducted in the Atten-ESN.

The result of RUL prediction based on Atten-ESN is given in Fig.4. Intuitively, the differences between the predicted RUL and ground truth RUL is slight. The MSE and Score of the prediction are 200.3 and 264.3 respectively.

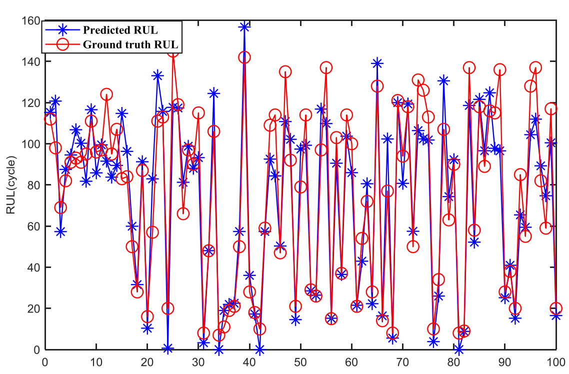

Figure 4. Prediction results of Atten-ESN model on FD001 with MSE=200.3, Score=264.3.

3.4 Comparison and Discussion

In this section, some comparison works are conducted to validate the effectiveness of the proposed Atten-ESN approach. For one thing, the comparisons of prediction performance between typical ESN with no attention and the Atten-ESN are performed to illustrate the validity of the additive attention mechanism in ESN. For another, some existing state-of-the-art prognostic approaches, such as random forest (RF), support vector machine (SVM), etc., and some deep learning approaches, e.g. deep belief network (DBN) [17] and long-short term memory (LSTM) [7], are utilized to benchmark with the proposed approach.

Fig.5 shows boxplot of prediction results regrading to MSE and score, both of which are attained with different ESN-based approach over 20 experimental runs. The position and height of the box can represent the prediction accuracy and stability respectively. As shown in Fig.5, the Atten-ESN with ridge regression has the better performance on both MSE and score in contrast to ESN approach with ridge regression. Besides, the Atten-ESN with ridge regression performs better than the ESN with linear regression on prediction accuracy. Moreover, both the Atten-ESN approaches with different regressions perform better on stability than the ESNs without attention. Table 3 reports the related quantitative performance in term of MSE and score. Accordingly, the attention mechanism integrated with ESN contributes to improving the accuracy and stability of RUL prediction.

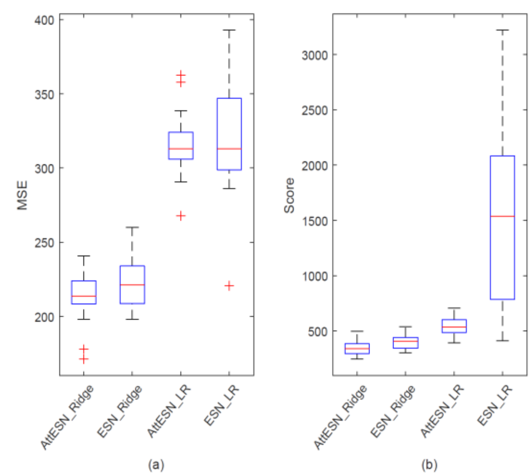

Figure 5. Comparisons of prediction on ESN-based approaches: (a) MSE of the RUL prediction; (b) Score of the RUL prediction.

Table 3. Comparisons of ESN-based approaches on FD001

Approaches	Regression	MSE	Score
Atten-ESN	Ridge	213.3	350.9
ESN	Ridge	223.1	406.8
Atten-ESN	Linear	315.8	540.9
ESN	Linear	320.1	1596.6

Additionally, the Atten-ESN with regression approach has been compared with some state-of-the-art approaches and the results are given in Table 4. As seen in the table, the proposed Atten-ESN consistently outperforms all of other compared approaches in terms of both MSE and score.

Table 4. Approaches Comparisons on FD001

Approaches	MSE	Score
DBN [17]	226.2	334.2
Deep LSTM [7]	260.5	338.0
Random Forest [17]	320.4	479.8
SVR [7]	439.3	1380.0
Atten-ESN	201.7	305.3

4. CONCLUSION

In this work we have developed a novel data-driven prognostic approach for RUL prediction. The proposed approach is based on ESN, which is a new type of RNN and has simple and effective training procedure. This paper develops a novel ESN structure via adding a simple yet effective attention mechanism to the standard ESN, which empowers the ESN to have the attentiveness capability and different importance levels of different elements in input vectors of a time slot can be adaptively treated. In addition, genetic algorithm is employed to optimize the parameters of the attention-based ESN and generalization ability of the proposed approach is improved. The effectiveness of the Atten-ESN approach is validated on the benchmark C-MAPSS dataset. The comparisons among different ESN-based approaches illustrate the validity of the additive attention model. Moreover, the comparison results demonstrate that the developed approach for RUL prediction has superior performance on both prediction accuracy and stability over other tested approaches. In the future research, the learning strategy of the attention related weights should be improved and thus the attention-based ESN can be extended to more applications.

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